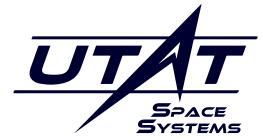


Electrical BUS Subsystem Literature Review



Overview of the Electrical BUS Subsystem

The Electrical BUS subsystem's goal is to provide the CubeSat with a centralized point of control, which at a high level will involve observing indicators and driving actuators. More specifically, Electrical BUS will interact with the PAYload, RF, and EPS subsystems to manage the state of affairs on the CubeSat. Operations of the electrical subsystem may include directing the PAYload subsystem to send data to the RF subsystem for down-linking, and acquiring uplink data from RF to perform a reboot.

Literature Review Methodology

As it pertains to the design, the main components that make up the electrical subsystem essentially boil down to three categories: control unit (that is, device(s) responsible for carrying out generic operations of the electrical subsystem, for example an MCU), protocol (that is, an interface for connecting two or more devices and allowing them to share data in one way or another, for example I2C), and peripherals (that is, device(s) responsible for a particular function within the Electrical BUS subsystem, for example a watchdog). This classification is not a recognized one, but has been found to be useful when conducting this literature review since the overwhelming majority of the designs can be deconstructed into components each of which would fit into one of these categories [subjective].

In terms of sources, two major categories are surveyed:

- Commercial OBCs discussed in NASA's Small-Sat State of the Art reports from 2022 to 2026 ([1, pp. 215–226], [2, pp. 225–237], [3, pp. 227–239], [4, pp. 241–253]), with trends also observed.
- OBCs built by design teams around Canada and beyond.

Design Space

This section outlines the resulting design space for the Electrical BUS Subsystem based on the corresponding literature review. We establish a space of common control units, protocols and peripherals for our internal design process.

Control Units

The control-unit design space ranges from low-power microcontrollers to heterogeneous processor-FPGA systems. NASA's State of the Art reports show that most conventional CubeSat OBCs use microcontrollers or embedded microprocessors because command execution, telemetry collection, timekeeping, fault management, and subsystem coordination do not require high computational performance [1, 2, 3, 4]. Higher-performance FPGAs, application processors, and system-on-chip devices are generally introduced when the OBC must also support image processing, autonomy, high-rate payload data, or machine-learning workloads.

Table 1: Candidate OBC control-unit architectures

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Single MCU or processor	One device performs command execution, telemetry, timekeeping, fault management, and subsystem control. Lowest complexity and generally lowest power.	EnduroSat OBC, ARM Cortex-M7 [2, p. 229]; AAC Clyde Space Kryten-M3, Cortex-M3 [2, p. 228].	STM32L4 PC/104 OBC; Athena TMS570; ORCASat TMS570 [5, 6, 7].
Multicore SoC	Multiple cores share one device and memory system. Provides additional performance or lockstep capability, but does not remove the SoC as a single point of failure.	Argotec FERMI, dual-core LEON3FT [2, p. 229]; SPiN MA61C, dual-core GR712RC [3, p. 236]; Xiphos Q7S, dual-core Cortex-A9 [3, p. 238].	Athena and ORCASat use lockstep-capable ARM safety microcontrollers [8, 9].
Dual redundant processors	Two processor devices provide cold, warm, or active redundancy. One processor may take control following a fault in the primary device.	Andes OBC, two Cortex-M7 processors with full cold redundancy [4, p. 230]; Bradford Binary OBC, dual Cortex-R5 with internal cold redundancy [2, p. 229]; Bradford Stellar OBC, dual PolarFire SoCs with internal cold redundancy [4, p. 230].	No directly equivalent two-device redundant architecture was identified in the academic survey.
Controller plus FPGA or DPU	A low-power controller performs critical spacecraft control while an FPGA or higher-performance processor handles payload data or acceleration. The secondary unit may be powered down when unused.	KP Labs Antelope, RM57 OBC plus Zynq UltraScale+ DPU [2, p. 230]; NOVI Gen 2, Vorago MCU plus Versal AI Edge SoC [4, p. 232].	NEUDOSE, primary AVR32 plus secondary Xilinx Zynq-7000 [10].
FPGA-centric	Control or processing is implemented using programmable logic and, optionally, a soft processor. Suitable for deterministic parallel interfaces and high-rate data paths, but more difficult to develop as a general-purpose OBC.	LASP Generic SmallSat FPGA Board, Kintex-7 [2, p. 230]; KP Labs Lion, Kintex UltraScale [2, p. 230].	No FPGA-only primary OBC was identified in the academic survey.

Table 1: Candidate OBC control-unit architectures (Continued)

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Distributed controllers	Subsystem-local processors control the EPS, ADCS, communications, or payload and exchange data with a coordinating OBC. This improves fault containment but increases network and software complexity.	NASA identifies federated, distributed, heterogeneous, and mixed-criticality avionics as viable SmallSat architectures [2, p. 225] [4, p. 221–222].	Individual subsystem controllers used alongside the primary academic OBC.

All architectures will be considered during preliminary design; however, the current baseline is a centralized microcontroller-based OBC with independent watchdog and reset circuitry. This approach is the most common among the academic systems surveyed and provides the lowest-risk balance of power, determinism, software complexity, and flight heritage. An FPGA or secondary processor may be added later if payload-processing requirements cannot be met by the primary MCU.

Protocols

Peripherals

COTS (Commercial Off the Shelf) Design Alternatives

Description: Determine what COTS solutions exist if we cannot produce a certain part of the subsystem in house. Find a COTS option for each in house design.

Will inform trades later on for COTS versus in house risks.

This section outlines alternative off-the-shelf components that would in part or in full fulfil the operating requirements of the subsystem.

TODO

System Architecture and Budgets

In this section we concern ourselves with the following considerations: power draw, physical dimentions, radiation tolerance, and weight for the purposes of the overall design and integration with other teams.

Power draw

Based on the commercial onboard-computing systems surveyed in NASA’s State of the Art reports [1, 2, 3, 4], OBC power consumption can be divided into three broad categories according to platform complexity and computational capability. The reports do not use a consistent average-power metric; reported values are variously identified as idle, static, typical, active-mode, full-load, or peak consumption. Consequently, Table 2 preserves the operating-state terminology used by the manufacturers.

The 2022 edition identifies representative onboard-computing products but does not tabulate their power consumption [1, Table 8-1, pp. 214–217]. Quantitative power values are therefore drawn primarily from the 2023, 2024, and 2026 editions [2, Table 8-1, pp. 228–233] [3, Table 8-1, pp. 230–236] [4, Table 8-4, pp. 229–234].

Table 2: OBC power-consumption trends and reported operating states

Platform Complexity	Reported Power Profile	Design Characteristics	Example Designs
Low (Bare-Metal / Control)	Minimum listed: 0.05 W Idle/steady: approximately 0.2–0.5 W Peak: generally below 1 W	Simple command and data handling, housekeeping, telemetry, and fault management. These systems are normally based on low-power microcontrollers and execute bare-metal software or a small real-time operating system. Their power consumption is relatively insensitive to computational load.	Pumpkin PPM family: 0.05 W, operating state not specified [2, p. 231] [3, p. 235] [4, p. 233]. AAC Clyde Space Kryten-M3: 0.4 W [2, p. 228] [3, p. 230] [4, Table 8-4]. C3S Electronics OBC: 0.46 W in test mode [2, p. 229] [3, p. 232] [4, p. 230]. SkyLabs NANOobc-2: < 0.9 W idle and < 1 W peak [2, p. 232] [3, p. 236].
Medium (Integrated OBC)	Idle: approximately 0.2–1 W Nominal/steady: approximately 1–3 W Peak: approximately 1.5–5 W	Typical integrated CubeSat OBCs combining command and data handling, mass-memory management, communication interfaces, fault protection, and basic payload management. Power may vary with processor utilization, interface activity, and memory access.	SPiN MA61C CubeSat: 1–1.2 W [2, p. 232] [3, p. 236] [4, Table 8-4]. AAC Clyde Space Sirius OBC & TCM: 1.3 W [2, p. 228] [3, p. 230] [4, Table 8-4]. EnduroSat OBC I: 0.2 W idle and 1.5 W peak [2, p. 229] [3, Table 8-1] [4, p. 230]. Resilient Computing RadPC: 1.5 W [2, p. 232] [3, p. 236] [4, p. 233]. SatRev OBC: 2.8 W typical and 5 W peak [4, p. 233].
High (Edge / AI / SoM)	Idle/static: approximately 5–15 W Typical/active: approximately 6–10 W Peak/full load: commonly 10–20 W, with larger payload computers reaching substantially higher values	Heterogeneous systems-on-module and payload-processing computers supporting image processing, software-defined radio, autonomous navigation, data compression, and machine-learning inference. Power is strongly dependent on processor utilization, FPGA configuration, accelerator activity, and payload duty cycle.	Novo Space GPU002AF: 6–11 W in active mode [2, p. 231] [3, p. 235] [4, p. 233]. KP Labs Leopard DPU: 7 W static and up to 10 W during deep-learning inference [2, p. 230] [3, p. 234] [4, p. 231]. Aitech S-A1760: ≤ 5 W idle, 8–10 W under a typical CUDA load, and < 20 W at full utilization [2, p. 228] [3, Table 8-1] [4, Table 8-4]. EnduroSat GPC: < 15 W idle and 130 W peak [4, p. 231].

Based on these results, a nominal power-design envelope of 1–10 W is adopted for the proposed OBC. This range encompasses conventional integrated CubeSat OBCs and moderate edge-processing workloads, while transient loads above 10 W are treated separately as peak operating conditions. The selected range is a project design boundary rather than the full commercial-hardware envelope, which extends from approximately 0.05 W for low-power microcontroller boards to 130 W peak for high-performance payload computers.

[TODO: peak loads and durations]

Dimentions

Weight

Radiation tolerance

Appendix A: Academic OBCs

Name	Manufacturer	MCU	Protocols	Standard	Notes	Ref
STM32L4 PC/104 academic OBC	VST104 thesis design	STM32L4—ARM-based	CAN	PC104	pc 104 cubesat design.	[5]
Athena OBC	University of Alberta (Albertasat)	TI TMS570LC4357 (ARM Cortex R5F)			Embedded memory type: Flash. Features ideal for satellite OBC application, including radiation & space reliability (rad-hard, SEL immune 48 MeV at 125°C, LAT 30krad), high computational performance (lockstep dual-core compare, 300MHz CPU clock, 1.66DMIPS/MHz).	[6, 11, 8]
Orcasat OBC	University of Victoria	TI TMS570LS0714 (Dual ARM Cortex-R4F)			Embedded memory type: Flash. Operates in lockstep, ECC integrated flash & RAM banks, radiation-tested up to 6 krad, 80MHz core clock.	[7, 9]
LORIS OBC	Dalhousie University	NXP i.MX6DL (Single-core ARM Cortex-A9)	USB 2.0; SPI; I2C; PWM; UART; GPIO; ADC		Embedded memory type: Flash. Low profile, power efficient SoC SBC. 4GB eMMC flash storage, 256MB DDR3 RAM, 800MHz operational frequency, -40°C to 85°C temperature range.	[12, 13]
NEUDOSE OBC	McMaster University	Primary: AVR32. Secondary: Xilinx Zynq-7000 SoC			Primary is COTS chosen for high TRL to reduce mission risk; secondary is designed for radiation tolerance using rad-hard and non-rad-hard components.	[10]
RADSAT-SK OBC	University of Saskatchewan	ARM926EJ-S			Embedded memory type: SDRAM. Uses ISISpace satellite-subsystems libraries, contains MMU, designed to interface with external SDRAM.	[14]

Appendix B: Table of commercial OBCs

Name	Manufacturer	MCU	Protocols	Standard	Notes	Ref
CFC-300	Innoflight	Xilinx Zynq SoC with dual-core ARM Cortex-A9 and FPGA fabric	CAN; I2C; RS422; RS485; SPI; SpaceWire		general-purpose; high performance. Fully on-orbit programmable; FPGA bit streams reloadable.	[15, 16], [1, p. 222], [2, p. 233], [3, p. 235], [4, p. 249]
CubeComputer V4.1	CubeSpace	32-bit ARM Cortex-M3 based MCU	CAN; I2C; SPI; UART	PC104	General-purpose CubeSat OBC; Piggyback header—mission specific expansion. Used as ADCS OBC on QB50 precursor satellites.	[17, 18]
Gen 2 OBC	NOVI Space	Vorago ARM Cortex M4		PC104	Embedded memory type: FRAM.	[1, p. 222], [2, p. 234], [3, p. 236], [4, p. 250]
iOBC	ISISPACE	Some Atmel ARM9 (32-bit ARM9) [datasheet]	GPIO; I2C; RS232; RS422; RS485; SPI; UART; USB			[19, 20]
Main Bus Unit SatBus 3C2	NanoAvionics		CAN; I2C; SPI; UART; USB		No datasheet; includes ADCS.	[21]

Name	Manufacturer	MCU	Protocols	Standard	Notes	Ref
NANOMIND A3200	GOMSPACE	Atmel AT32UC3C MCU / AVR32 MCU	CAN; Cubesat Space Protocol; I2C; SPI; UART		Tiny.	[22, 23], [1, p. 223], [2, p. 234], [3, p. 236], [4, p. 250]
NANOMIND HP MK3	GOMSPACE	Xilinx Zynq 7030 (FPGA + ARM) [product page]	CAN; RS422; RS485; SPI; SpaceWire; USB		High performance (ARM dual core); FPGA included. Flew on SDR MK III; has PPS.	[24, 25], [1, p. 225], [2, p. 236], [3, p. 238], [4, p. 252]
NanoMind HP (PolarFire)	GOMSPACE	32-bit RISC-V core / PolarFire		PC104	Embedded memory type: DDR3.	[1, p. 225], [2, p. 236], [3, p. 238], [4, p. 252]
OBC - P4	Space Inventor	ARM Cortex-M7	CAN		four OBC modules; hot/cold redundancy solutions (Datasheet: request on sat-search). FreeRTOS(tm) real time operating system.	[26]

Name	Manufac-turer	MCU	Protocols	Stan-dard	Notes	Ref
Sirius OBC LEON3FT	AAA Clyde Space	32-bit LEON3FT (IEEE-1754 SPARC v8) fault-tolerant pro- cessor	CAN; GPIO; RS422; RS485; SpaceWire	SpaceWire 50 Mbps	Reliable high-performance for advanced missions; designed and qualified for five years in LEO. Embedded memory type: NAND Flash.	[27, 28], [1, p. 219], [2, p. 230], [3, p. 232], [4, p. 246]
[the] ON- BOARD COM- PUTER	EnduroSat	ARM Cortex M7	I2C	PC104	Option to include GNSS receiver (Datasheet: [reg walled]). Embedded memory type: Flash.	[29], [1, p. 222], [2, p. 233], [3, p. 235], [4, p. 249]

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